2.10.11

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Question:
Construct a triangle ABC with side $\mathbf{BC} = 6cm, \angle B = 45, \angle A = 105$
Solution:
Let position vector of **B** be $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and a,b,c be the sides opposite to the vertices A,B and C
Given a=6cm;

$$\mathbf{C} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \tag{0.1}$$

$$\mathbf{A} = \begin{pmatrix} c\cos\angle B \\ c\sin\angle B \end{pmatrix} \tag{0.2}$$

$$\mathbf{A} = \begin{pmatrix} \frac{c}{\sqrt{2}} \\ \frac{c}{\sqrt{2}} \end{pmatrix} \tag{0.3}$$

$$\angle A + \angle B + \angle C = 180 \tag{0.4}$$

$$\therefore \angle C = 30 \tag{0.5}$$

in ΔABC

$$b\cos\angle C + c\cos\angle B = a \tag{0.6}$$

$$b\sin\angle C - c\sin\angle B = 0 \tag{0.7}$$

$$\begin{pmatrix} \cos\angle C & \cos\angle B \\ \sin\angle C & -\sin\angle B \end{pmatrix} \begin{pmatrix} b \\ c \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \tag{0.8}$$

using augmented matrix

$$\left(\begin{array}{cc|c} \cos\angle C & \cos\angle B & 6 \\ \sin\angle C & -\sin\angle B & 0 \end{array}\right) \tag{0.9}$$

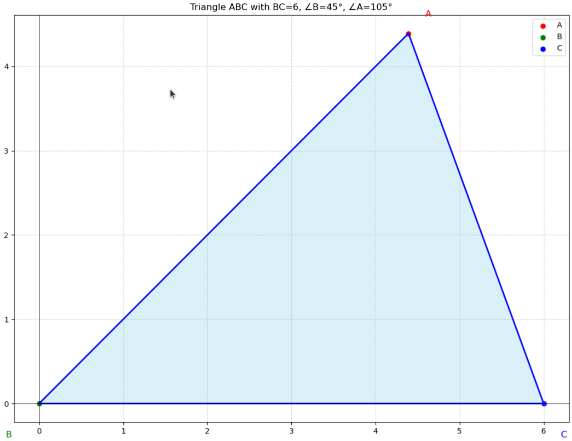
$$\left(\begin{array}{cc|c} \sqrt{3}/2 & 1/\sqrt{2} & 6 \\ 1/2 & -1/\sqrt{2} & 0 \end{array}\right) \tag{0.10}$$

$R_2 \longrightarrow \sqrt{3}R_2 - R_1$

$$\left(\begin{array}{cc|c} \sqrt{3}/2 & 1/\sqrt{2} & 6 \\ 0 & -\sqrt{3} - 1/\sqrt{2} & -6 \end{array}\right) \tag{0.11}$$

$$\implies c = \frac{6\sqrt{2}}{\sqrt{3}+1} \tag{0.12}$$

$$\therefore \mathbf{A} = \begin{pmatrix} 6\left(\sqrt{3}-1\right) \\ 6\left(\sqrt{3}-1\right) \end{pmatrix} \tag{0.13}$$


Triangle ABC with BC=6, ∠B=45°, ∠A=105°
A
B
C
0
1
2
3
4
5
6